

Course Introduction. Revision of statistical modelling. Main concepts in AI.

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Class 1



- 1 Motivation
- 2 Course introducton
- 3 Linear regression
- 4 Gradient descent
- 5 Logistic regression
- 6 Regularization
- 7 Conclusion

Learning outcomes of today's class

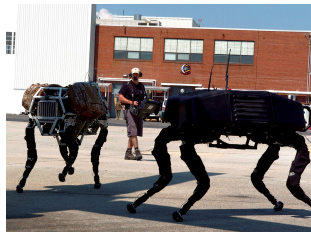
By the end of today's class, you should

- ① Get excited by various application of AI in the finance industry
- ② Understand basic terms and concepts in AI
- ③ Review statistical modelling

Section 1

Motivation

What is AI?

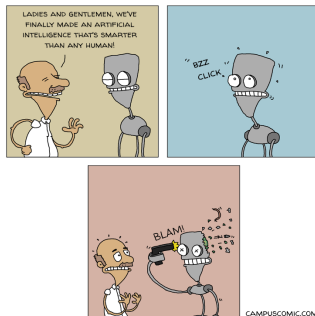


Course Aim

Building on Programming and Probability / Statistics / Linear Financial Models this course aims to sparkle your interest to Artificial Intelligence (AI) and to equip you with basic technical knowledge of modern machine learning techniques that will help you to tell hype from real things in the area of AI.

Problem

What is real? What is real in the near future? What is real but too costly?



Activity: ask ChatGPT to give you career advice. Experiment with it, ask it to give you a one year plan for how you can achieve your career goals, tell it that it can ask you for more information if needed etc. Then we'll discuss

Fedor's prompt: "I am 45 years old and I have a Ph.D. in algebraic topology. I have worked as a mathematics lecturer at Nanyang Technological University, Singapore, for 17 years. I have taught courses such as calculus, linear algebra, topology, differential equations, data science, artificial intelligence, programming, and network science. I have published papers in pure mathematics, education, chemistry, data science, and game theory. I am now thinking of leaving academics and changing my career. Can you give me 10 possible positions that I can look for?"

- <https://app.wooclap.com/ETBFFC>



Figure 1: WooClap activity

Major trends in business affect finance, investment, trading, risk management etc.

Losing their jobs to bots

Autonomous Research estimates that 1.2 million people working in banking and lending will be replaced by artificial intelligence software by 2030

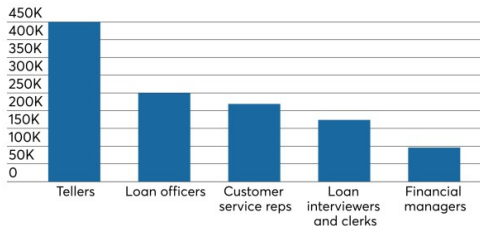


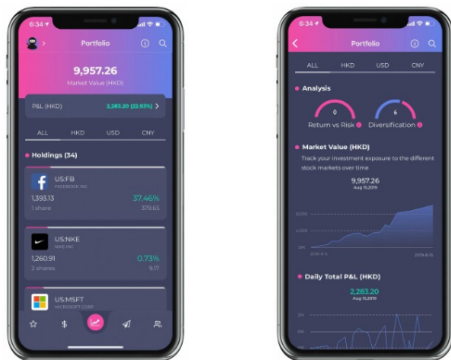
Figure 2: Loss of banking jobs due to AI

<https://www.americanbanker.com/news/how-artificial-intelligence-is-reshaping-jobs-in-banking>

Companies doing AI in finance

8 securities: financial robo-advisor “Chloe”:

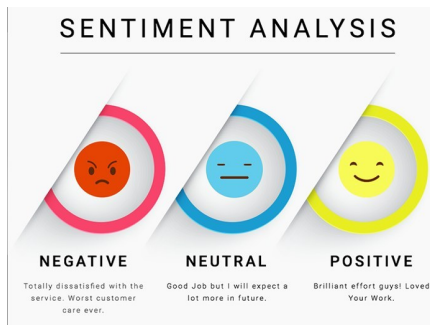
Good investing starts with a goal. You start by telling Chloe a little bit about yourself including your age, income, and goal. You can start with as little as HK\$ 1,000. Chloe invests your money into a globally diversified portfolio of exchange traded funds. Chloe monitors your goals and optimizes your portfolio for you. You can invest more or withdraw your money at any time with no penalty or questions asked.



RavenPack: market research

RavenPack is the leading big data analytics provider for financial services. Financial professionals rely on RavenPack for its speed and accuracy in analyzing large amounts of unstructured content. The company's products allow clients to enhance returns, reduce risk and increase efficiency by systematically incorporating the effects of public information in their models or workflows. RavenPack's clients include the most successful hedge funds, banks, and asset managers in the world.

Specific application: sentiment analysis for predicting buyouts.



Taiger: knowledge extraction and summarization.

Read and identify key information across multiple documents and databases.
Automate data validation & cleaning. Taiger Extract is an award winning document extraction tool built for both structured data and unstructured information with industry leading accuracy.

Nomura invests in Brevan Howard AI trading platform. Brevan Howard developed the AIM² platform three years ago using artificial intelligence, machine learning and data science.

<https://www.thetradenews.com/nomura-invests-brevan-howard-ai-trading-platform/>

Section 2

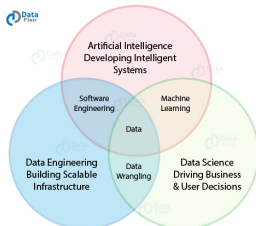
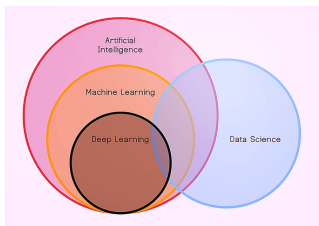
Course introduciton

Main terms

Colloquially, the term “artificial intelligence” is often used to describe machines (or computers) that mimic “cognitive” functions that humans associate with the human mind, such as “learning” and “problem solving” (Wikipedia).

Machine learning (ML) is the scientific study of algorithms and statistical models that computer systems use to perform a specific task without using explicit instructions, relying on patterns and inference instead. (Wikipedia).

Data science is a multi-disciplinary field that uses scientific methods, processes, algorithms and systems to extract knowledge and insights from structured and unstructured data. (Wikipedia).



Learning outcomes of the whole course

Upon successful completion of this course, students should be able to:

- ① Implement AI models in Python / Tensorflow / Keras.
- ② Evaluate applicability of AI models to solve financial problems.
- ③ Apply AI techniques to a real life problem in finance and evaluate the quality of solution in terms of model accuracy, interpretability, efficiency, scalability, and cost of collecting data.
- ④ Work in a team, i.e., effectively communicate on distribution of roles and work tasks and help other students with their tasks.
- ⑤ Explain AI models in the form of a report written at the appropriate level for the intended audience.

Assessment

- Two individual quizzes 20% (weeks 2 & 4)
- Two team assignments 20% (weeks 3 & 5)
- Programming test 20% (week 6)
- Group project 40% (week 7)

Supervised machine learning

Example: <https://www.kaggle.com/uciml/default-of-credit-card-clients-dataset>

LIMIT_BAL	SEX	EDUCATION	AGE	PAY_0	PAY_2	default
20000	2	2	24	2	2	yes
120000	2	2	26	-1	2	yes
90000	2	2	34	0	0	no
50000	2	2	37	0	0	no
50000	1	2	57	-1	0	no
50000	1	1	37	0	0	no

Features / predictors / independent variables:

LIMIT_BAL SEX EDUCATION MARRIAGE AGE PAY_0 PAY_2 PAY_3 PAY_4
PAY_5 PAY_6 BILL_AMT1 BILL_AMT2 BILL_AMT3 BILL_AMT4 BILL_AMT5
BILL_AMT6 PAY_AMT1 PAY_AMT2 PAY_AMT3 PAY_AMT4 PAY_AMT5
PAY_AMT6

Dependent variable / response variable / output

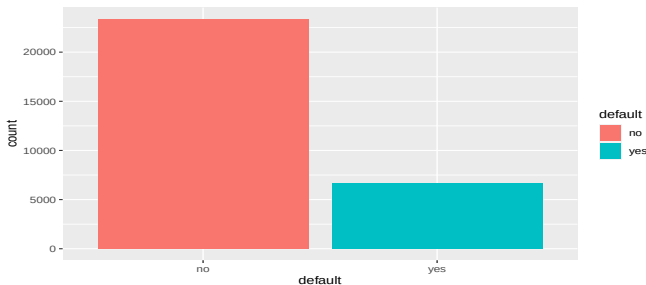
default

Problem

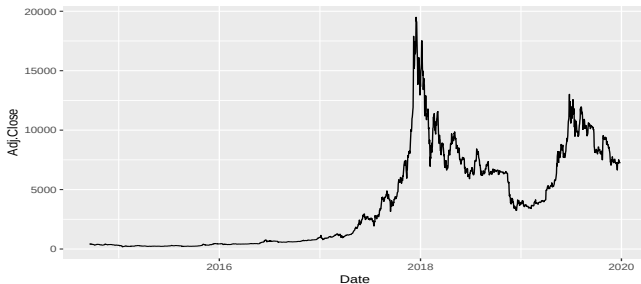
Prediction: given a new customer, predict if he is going to default.

LIMIT_BAL	SEX	EDUCATION	AGE	PAY_0	PAY_2	default
420000	2	1	33	-1	-1	?
180000	2	2	40	0	-1	?
260000	1	2	28	0	0	?

Challenge: imbalance of data. False positives vs false negatives.



Example: <https://sg.finance.yahoo.com/quote/BTC-USD?p=BTC-USD>



Reformatting the data

d_5	d_4	d_3	d_2	d_1	d_0
457.33	424.44	394.80	408.90	398.82	402.15
424.44	394.80	408.90	398.82	402.15	435.79
394.80	408.90	398.82	402.15	435.79	423.20
408.90	398.82	402.15	435.79	423.20	411.57
398.82	402.15	435.79	423.20	411.57	404.42
402.15	435.79	423.20	411.57	404.42	399.52

Problem

Given recent history, predict the price of bitcoin tomorrow.

	d_5	d_4	d_3	d_2	d_1	d_0
1000	2659.63	2717.02	2506.37	2464.58	2518.56	?

Features / predictors / independent variables: d_5 d_4 d_3 d_2 d_1

Dependent variable / response variable / output

d_0

Challenge: time series. Predictors are not independent.

Classification vs regression

Classification — the response variable is categorical. Output of a machine learning algorithm is a probability vector. Below is a credit card default example.

LIMIT_BAL	SEX	EDUCATION	AGE	PAY_0	PAY_2	yes	no
420000	2	1	33	-1	-1	0.14	0.86
180000	2	2	40	0	-1	0.11	0.89
260000	1	2	28	0	0	0.16	0.84

Accuracy of classification is measured from the confusion matrix.

Prediction	Reference	
	no	yes
no	22582	5560
yes	782	1076

Regression — the response variable is numeric. Output of a machine learning algorithm is a number. Below is a bitcoin example

	d_5	d_4	d_3	d_2	d_1	d_0_predicted
1000	2659.63	2717.02	2506.37	2464.58	2518.56	2523.94

Accuracy of regression — root-mean-square error or mean absolute error.

More complex applications that require advanced models:

Text processing, e.g., sentiment of twitter messages.

Example from <https://www.kaggle.com/davidwallach/financial-tweets>: “The price of lumber \$LB_F is down 22% since hitting its YTD highs. The Macy’s \$M turnaround is still happening.. <https://t.co/XnKsV4De39>”

Image processing. Below is an example



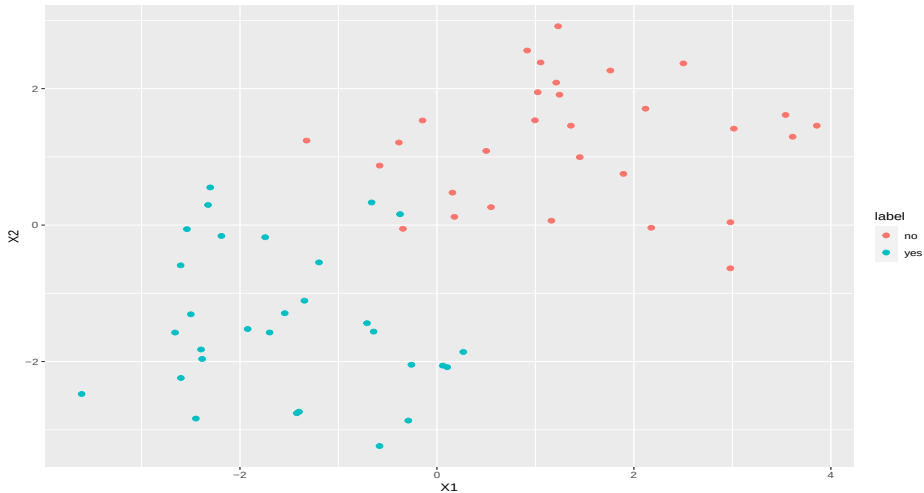
Unsupervised learning

In supervised learning, we have a labelled dataset and the aim is to guess the pattern and to come up with some algorithm that will label new data.

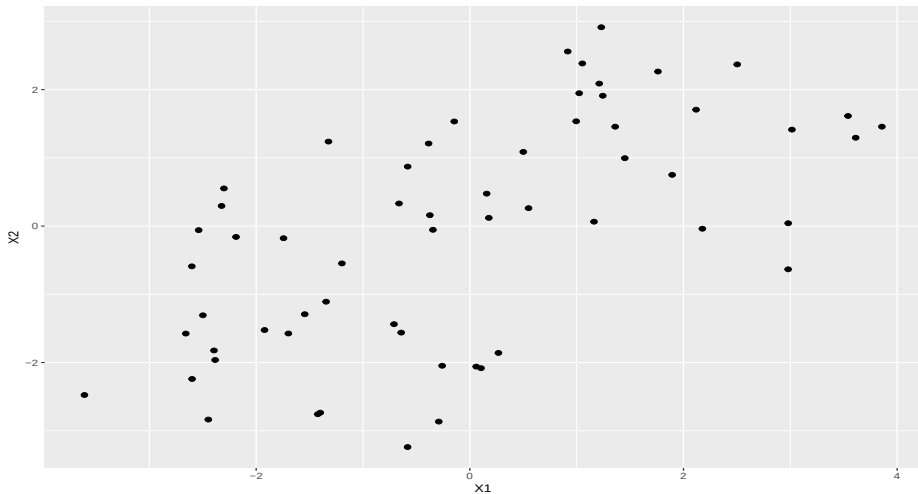
In unsupervised learning, we have an unlabelled dataset and the aim is to discover any useful information about it.



Supervised learning



Unsupervised learning



Section 3

Linear regression

Definition

Features (predictors): $X_1, \dots, X_p$. Or, in vector form, the input is

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{bmatrix}$$

Quantitative output Y .

Training data:

$$(x^1, y^1), (x^2, y^2), \dots, (x^N, y^N)$$

Training data in the matrix form:

$$\mathbf{y} = \begin{bmatrix} y^1 \\ y^2 \\ \vdots \\ y^N \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_1^1 & \cdots & x_p^1 \\ 1 & x_1^2 & \cdots & x_p^2 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_1^N & \cdots & x_p^N \end{bmatrix}$$

We assume that the true model is

$$E[Y|X] = f(X) = \beta_0 + \sum_{j=1}^p X_j \beta_j$$

or, equivalently,

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon,$$

where ε is the error term (random noise). Its expectation is 0.

We do not know the true values of $\beta_0, \beta_1, \dots, \beta_p$. The best thing we can do is to find parameter values that are most consistent with the observed data.

Loss function

Suppose the true model is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon$$

If we have some estimates $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ for the regression parameters, then we can make predictions by

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p$$

Values $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ for the model parameters are chosen as to minimize the mean squared error (ordinary least squares) *loss function*

$$L(\beta) = \sum_{i=1}^N (y^i - \hat{y}^i)^2 = \sum_{i=1}^N \left(y^i - \beta_0 - \sum_{j=1}^p x_j^i \beta_j \right)^2$$

How to minimize the loss function for linear regression.

Activity: property prices

We will review basics of linear regression and think about some important concepts and pitfalls in machine learning.

Data: <https://data.gov.sg/dataset/resale-flat-prices>

Material: notebook 1



- <https://app.wooclap.com/ETBFFC>



Figure 3: WooClap activity

Takeaways:

- ① The process of constructing a model is referred to as “training”. The dataset used for training is called “training data”.
- ② Goodness of fit for regression models is usually estimated by mean absolute error while training is done via minimizing the mean squared error loss function.
- ③ To get an estimate of how good a model is, we apply it to a “test data” - a dataset that the model has not seen before but for which correct answers are known.

- ④ Of course, the more predictors we have, the more precise model we will get. In fact, even if we have 1000 observations of 999 completely random independent variables with a completely random independent outcome, the linear model with 999 regressors will have 1000 degrees of freedom, i.e., it will fit the training data perfectly.
- ⑤ In statistics, overfitting is “the production of an analysis that corresponds too closely or exactly to a particular set of data, and may therefore fail to fit additional data or predict future observations reliably” (Wikipedia).

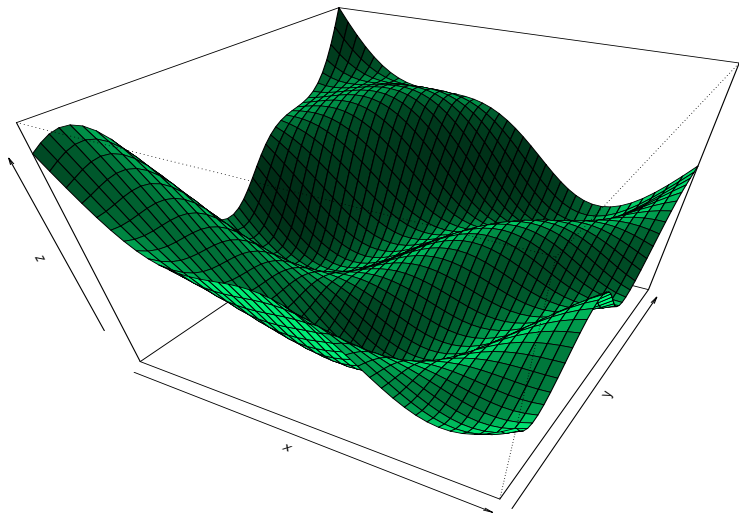
Section 4

Gradient descent

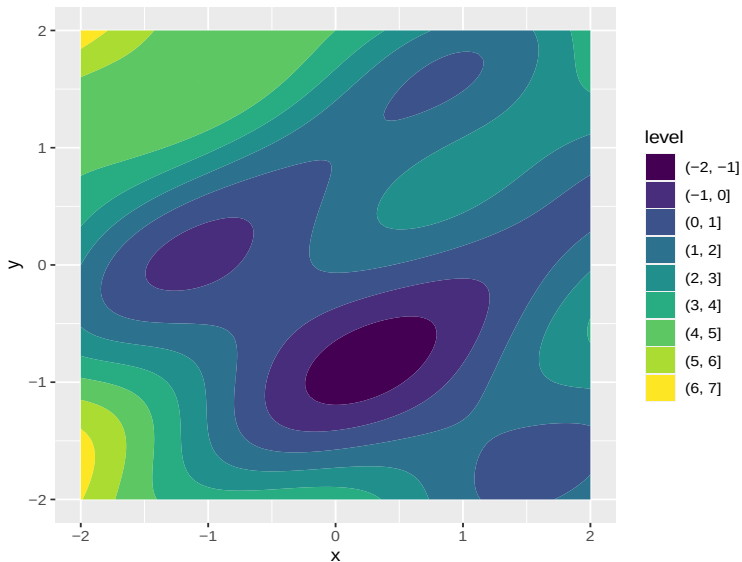
Gradient descent

The graph of a function $z = f(x, y)$ is a surface in the 3D space.

$$z = f(x, y)$$



Another way to visualize a function $z = f(x, y)$ is a contour plot:



The *gradient* of a function $f(x, y)$ is the vector

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$$

The gradient is orthogonal to level curves of the function and points at the direction in which the function increases at the fastest rate.

Gradient descent algorithm:

- Begin at some initial point (x_0, y_0) and find the gradient vector $\nabla f(x_0, y_0)$.
- Make a small step in the direction $-\nabla f$, i.e.,

$$(x_1, y_1) = (x_0, y_0) - \alpha \nabla f(x_0, y_0)$$

- Make a small step in the direction $-\nabla f$ again:

$$(x_2, y_2) = (x_1, y_1) - \alpha \nabla f(x_1, y_1)$$

- Etc. Either stop after a fixed number of steps or when $\nabla f \approx 0$.

Section 5

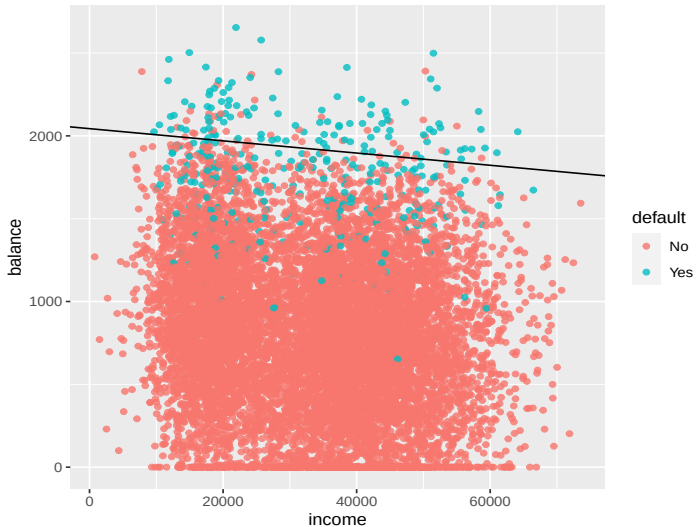
Logistic regression

Linear decision boundary

Example: credit card default data (from “Introduction to Statistical Learning”)



Linear decision boundary:



A linear decision boundary for binary classification can be constructed as follows. First, we change two classes into zeroes and ones:

student	balance	income	Y
No	549.19	43688.68	0
Yes	1581.07	17331.33	0
Yes	455.70	19080.81	0
Yes	1118.70	21848.44	1
Yes	991.23	8926.80	0
Yes	1255.85	22022.58	0
No	1021.28	44718.41	0
Yes	1334.97	14834.86	0
No	538.16	44654.09	0
No	766.51	43426.83	0

Then we fit a linear regression $\hat{Y} = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p = l(X)$.

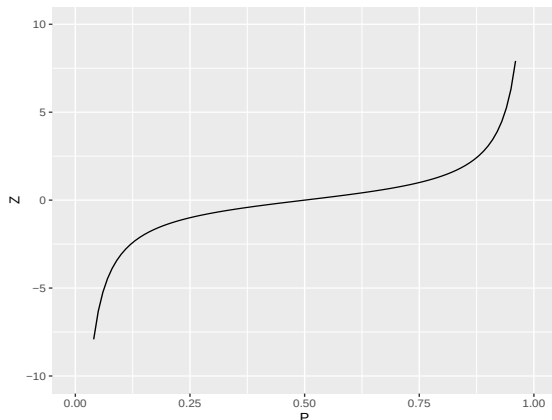
The classifier is

$$C(x) = \begin{cases} 1, & \text{if } l(x) > 0.5 \\ 0, & \text{if } l(x) < 0.5 \end{cases}$$

Logit transformation

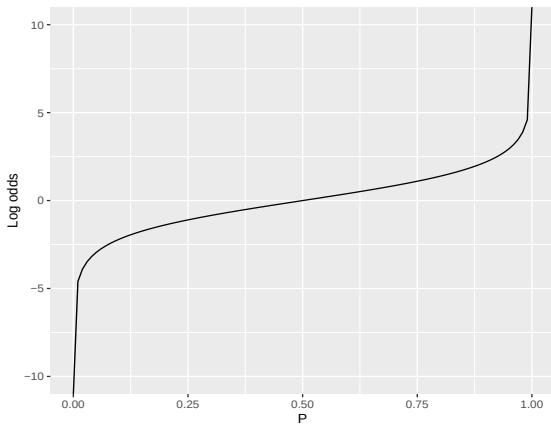
In classification problems, we want to predict the probability rather than just labels. And we have $0 \leq P \leq 1$ while there is no such restriction on $\hat{Y}$.

Instead of P , we can predict $Z = \varphi(P)$ that takes values from $-\infty$ to ∞



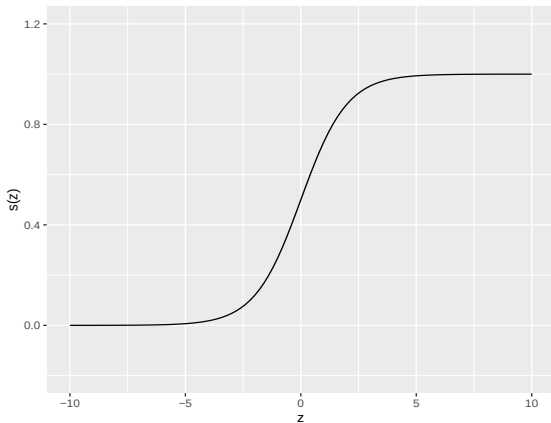
Log odds or logit transformation is

$$\log \frac{P}{1-P}$$



Sigmoid function

$$s(z) = \frac{e^z}{1 + e^z} = \frac{1}{1 + e^{-z}}$$



Logistic regression

Recall that we have two outcomes, 0 and 1 and the outcome of our classification model is a vector $(\hat{y}_0, \hat{y}_1)$. Then let $\hat{y}^i = \hat{y}_1^i$ be the estimated probability that $Y = 1$ given that $X = x^i$. More generally, let $p(X)$ be the estimated probability

$$P(Y = 1|X)$$

Logistic regression is actually a model for binary classification:

$$\log \frac{p(X)}{1 - p(X)} = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$$

The likelihood of observing outcomes as they are is

$$l(\beta) = \prod_{i:y^i=1} p(x^i) \times \prod_{i:y^i=0} (1 - p(x^i))$$

We choose the value of β maximizing $l(\beta)$. But maximizing $l(\beta)$ is the same as maximizing $\log l(\beta)$.

Derive the loss function $L(\beta) = -\log l(\beta)$.

Demo: <https://www.desmos.com/calculator/2gpbfbukht>

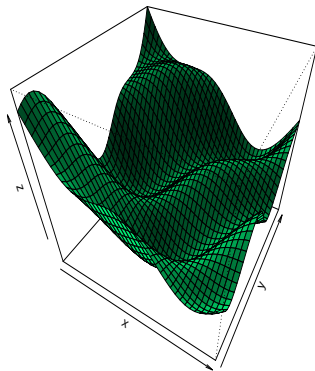
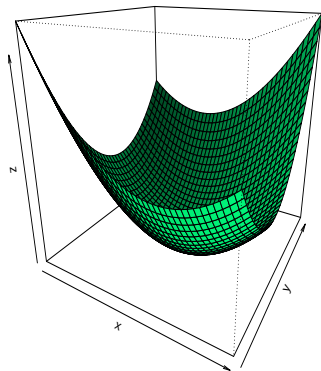
Definition

The logistic regression is the model

$$p(X) = s(X^T \beta) = \frac{1}{1 + e^{-X^T \beta}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p)}}$$

The loss function is the so-called *binary cross-entropy*

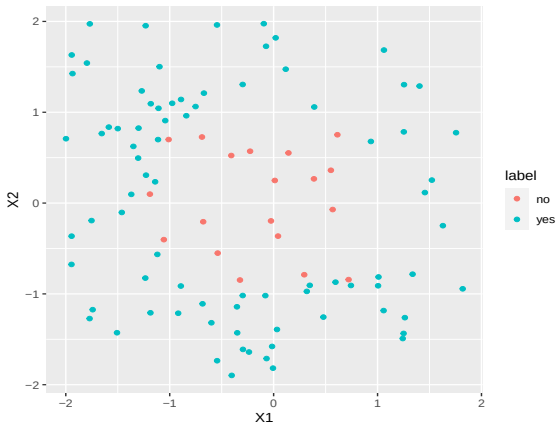
$$L(\beta) = - \sum_{i=1}^N (y^i \log p(x^i) + (1 - y^i) \log(1 - p(x^i)))$$



Logistic regression is trained by some version of gradient descent. The loss function is, in fact, convex. There is a unique point of minimum.

If the true decision boundary is non-linear, we may need powers of predictors as new predictors, i.e., combine the sigmoid function with a polynomial model.

Example:



Section 6

Regularization

Main idea

When a statistical model has too many parameters, it may be prone to overfitting. A popular way to deal with it is *regularization*.

The usual linear regression (for N observations and p predictors) is

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

with the mean squared error loss

$$L(\beta_0, \beta_1, \dots, \beta_p) = \frac{1}{N} \sum_{i=1}^N \left(y^i - (\beta_0 + \beta_1 x_1^i + \cdots + \beta_p x_p^i) \right)^2$$

To train the model, we choose values of $\beta_0, \beta_1, \dots, \beta_p$ to minimize the loss

Regularization is modifying the loss function by adding a term that “penalizes” the model for being too complex.

Lasso regression

Lasso regression is linear / logistic regression with the loss function modified by including the L_1 -regularization term as follows:

$$L(\beta_0, \beta_1, \dots, \beta_p) = \frac{1}{N} \sum_{i=1}^N \left(y^i - (\beta_0 + \beta_1 x_1^i + \dots + \beta_p x_p^i) \right)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

Here, λ is a *hyperparameter*. It governs the tradeoff between goodness of fit and complexity of the model.

Other types of regularization: L_2 regularization (with squares instead of absolute values), elastic net (combination of L_2 and L_1), more complicated types.

Depending on the value of λ , some predictors may disappear.

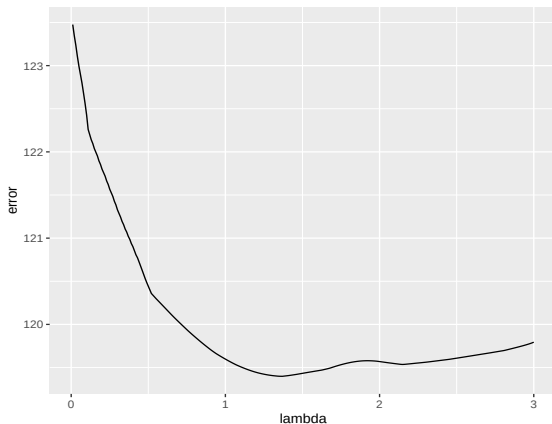
Here is an example where we predict the price of bitcoin (not the movement!) based on the price of bitcoin on 10 previous days.

	1000	100	10	1	0.1	0.01
(Intercept)	964.712	108.054	22.035	13.448	13.176	13.160
d_10	.	.	.	.	-0.012	-0.013
d_9	.	.	.	.	0.002	0.004
d_8	.	.	0.003	0.003	0.011	0.015
d_7	.	.	.	.	-0.006	-0.010
d_6	.	.	.	.	-0.018	-0.029
d_5	.	.	0.01	0.011	0.034	0.047
d_4	.	.	.	.	-0.009	-0.017
d_3	.	.	.	.	0.001	0.002
d_2	.	0.008	0.005	0.004	-0.001	-0.005
d_1	0.745	0.964	0.977	0.979	0.995	1.003

Optimal value of the hyperparameter

The optimal value of the hyperparameter λ is found by experimenting rather than via training. It is basically trial and error.

We created training and test sets. Then for $\lambda = 0.01, 0.02, \dots, 3$, we trained the lasso regression on the *training* set to predict the bitcoin price based on 30 previous days. Below is the *test* mean absolute error for each value of λ :



Training, validation, and test sets

Here is how the method works: we first choose some values

$$\lambda = l_1, l_2, \dots, l_m$$

Then, for each of these values, we train the lasso model on the training set and calculate its test error. Then we pick the value of λ that yields the smallest test error.

- <https://app.wooclap.com/ETBFFC>



Figure 4: WooClap activity

Discuss why this method is incorrect (WooClap).

A better method is splitting all available data into a training set, a validation set, and a test set. Then, for each value of λ , we train the lasso regression on the training set and calculate its error on the validation set. Then we pick the value of λ that yields the smallest validation error and retrain the lasso regression on the training and validation sets combined. Finally, in order to get an independent estimate of the error, we calculate the error on the test set.

Another very common approach is *k-fold cross-validation*. It is used when the dataset is small and the ML model doesn't take too long to train. It is recommended to read about cross-validation, but it won't be tested in this class.

Section 7

Conclusion

Checklist of concepts

- 1 Artificial intelligence vs machine learning vs data science.
- 2 Supervised vs unsupervised learning.
- 3 Regression vs classification.
- 4 What it means to train a model. Loss function. Parameters vs hyperparameters.
- 5 Classical parametric statistical models. Significance of coefficients.
- 6 Overfitting.
- 7 Training, validation, and test datasets.

Checklist of main learning points

- ① It is important that a predictive model is applied on the same type of data as data it was trained on.
- ② Models with a lot of parameters are prone to overfitting.
- ③ One of possible methods to deal with overfitting is regularization. It involves setting a value for a hyperparameter λ . Choosing the optimal value of λ requires some kind of trial and error.

Homework

Notebook 2: Logistic regression that predicts whether the price of bitcoin will go up or down based on historical prices. Notebook 2 contains questions for discussion in the beginning of the next class and theoretical questions.

Optional: Add the location of an HDB block to model 4 in Notebook 1 as one of predictors. Will such an improved model still produce nonsensical predictions of future HDB prices?

